

UGB Mock-2: Solutions

Problem 1

Part (a). Define $F(x) = \int_0^x f(t) dt$. Since f is continuous, F is differentiable with $F'(x) = f(x)$. The first given condition states that $F(1) = 0$, and trivially $F(0) = 0$.

Apply integration by parts to the second condition:

$$0 = \int_0^1 x f(x) dx = [xF(x)]_0^1 - \int_0^1 F(x) dx.$$

Since $F(1) = 0$, this reduces to $\int_0^1 F(x) dx = 0$. By the Mean Value Theorem for integrals, the continuity of F and the vanishing of its integral over $[0, 1]$ forces the existence of some $r \in (0, 1)$ with $F(r) = 0$.

We now have three distinct points where F vanishes: $F(0) = F(r) = F(1) = 0$. Rolle's Theorem applied on $[0, r]$ and on $[r, 1]$ produces $c_1 \in (0, r)$ and $c_2 \in (r, 1)$ with $F'(c_1) = F'(c_2) = 0$. Since $F' = f$, we have $0 < c_1 < c_2 < 1$ with $f(c_1) = f(c_2) = 0$.

Part (b). Consider the quadratic $Q(x) = x^2 - x + \frac{3}{16}$. Pairing it against f and using all three given conditions gives

$$\int_0^1 Q(x)f(x) dx = \int_0^1 x^2 f(x) dx - \int_0^1 x f(x) dx + \frac{3}{16} \int_0^1 f(x) dx = 1.$$

Next, compute the L^1 norm of Q on $[0, 1]$. Its roots are $x = 1/4$ and $x = 3/4$, so

$$\int_0^1 |Q(x)| dx = \int_0^{1/4} Q + \int_{1/4}^{3/4} (-Q) + \int_{3/4}^1 Q = \frac{1}{16}.$$

Suppose, for contradiction, that $|f(x)| \leq 16$ throughout $[0, 1]$. Then

$$1 = \int_0^1 Q(x)f(x) dx \leq \int_0^1 |Q(x)| \cdot |f(x)| dx \leq 16 \int_0^1 |Q(x)| dx = 16 \cdot \frac{1}{16} = 1.$$

Equality must therefore hold throughout the chain. The second inequality forces $|f(x)| = 16$ almost everywhere, and the first forces $f(x) = 16 \cdot \text{sgn}(Q(x))$ wherever $Q(x) \neq 0$. But Q changes sign at $x = 1/4$ and $x = 3/4$, so f would have to jump between $+16$ and -16 at these points, contradicting the continuity of f . Hence some $c \in (0, 1)$ must satisfy $|f(c)| > 16$.

Problem 2

Let p be an odd prime and consider $x^n + y^n = p^{n-1}(x + y)$ with $\gcd(x, y) = 1$ and $n \geq 2$.

If n is even, then $x + y$ does not divide $x^n + y^n$ as a polynomial, so we examine the prime structure directly. Suppose $p \mid x + y$. Then $x \equiv -y \pmod{p}$, so $x^n + y^n \equiv 2x^n \pmod{p}$. For p to divide

$x^n + y^n$, it would then have to divide x , and hence y , violating $\gcd(x, y) = 1$. Therefore n must be odd.

For odd n , set $Q = \frac{x^n + y^n}{x + y} = p^{n-1}$. If $p \mid x + y$, the Lifting the Exponent lemma gives

$$v_p(x^n + y^n) = v_p(x + y) + v_p(n),$$

so $v_p(Q) = v_p(n)$. Since $Q = p^{n-1}$, we need $n - 1 = v_p(n)$. But $v_p(n) \leq \log_p n$, which leaves no solutions for $n \geq 2$. Hence $p \nmid x + y$.

By Zsigmondy's theorem, for $n \geq 3$ the number $x^n + y^n$ has a primitive prime divisor q with $q \equiv 1 \pmod{2n}$. Because $Q = p^{n-1}$, this primitive prime is p itself, so $p \geq 2n + 1$. The algebraic lower bound $Q \geq \frac{(x + y)^{n-1}}{2^{n-1}}$ rearranges to $x + y \leq 2p$, and for $n \geq 5$ the term p^{n-1} outstrips every allowable bound on $x + y$. This leaves only $n = 3$.

For $n = 3$, the equation simplifies to $x^2 - xy + y^2 = p^2$, which is the norm form of the Eisenstein integers $\mathbb{Z}[\omega]$. Since $p \equiv 1 \pmod{6}$, it splits as $p = a^2 - ab + b^2$ for some positive integers $a > b$. Squaring in $\mathbb{Z}[\omega]$ gives the parametrization $(x, y) = (a^2 - b^2, 2ab - b^2)$ together with its symmetric variants.

Hence the only solutions are $n = 3$ with (x, y) parametrized as above, and only for primes $p \equiv 1 \pmod{6}$.

Problem 3

Part (a). We count sequences $\epsilon \in \{-1, +1\}^n$ whose partial sums $S_k = \sum_{i=1}^k \epsilon_i$ satisfy $|S_k| \leq k$ for every $1 \leq k \leq n$. This is precisely the count of simple lattice paths of length n that remain inside the symmetric strip $[-k, k]$.

The classical Reflection Principle handles the two-barrier problem by alternately reflecting across the barriers $\pm(k + 1)$, producing an inclusion-exclusion over shifts of size $2(k + 1)$. Summing over all permissible final heights $S_n \in [-k, k]$, the count collapses to

$$\sum_{j \in \mathbb{Z}} (-1)^j \binom{n}{\lfloor n/2 \rfloor + j(k + 1)},$$

as required.

Part (b). We must show that at least half of the 2^n sequences satisfy $M(\epsilon) \geq \lfloor \sqrt{n} \rfloor$, equivalently, that the number of sequences with $M(\epsilon) \leq \lfloor \sqrt{n} \rfloor - 1$ does not exceed 2^{n-1} .

Set $k = \lfloor \sqrt{n} \rfloor - 1$ in part (a). The quantity to bound is

$$N = \sum_{j \in \mathbb{Z}} (-1)^j \binom{n}{\lfloor n/2 \rfloor + j \lfloor \sqrt{n} \rfloor}.$$

This sum is exactly the number of simple random walks of length n that never leave the strip $[-\sqrt{n}, \sqrt{n}]$, a window of roughly one standard deviation. Standard binomial tail estimates (for instance, Hoeffding's inequality, or directly the central limit theorem) show that the probability of such confinement is strictly less than $1/2$, so $N \leq 2^{n-1}$.

Problem 4

Let ABC be a scalene triangle with circumcircle Ω , incenter I , and let the line AI meet Ω again at M . By the Incenter-Excenter Lemma, M is equidistant from B , C , I , and the A -excenter I_A . Thus the circle centered at M through B and C also contains I and I_A .

This circle meets the line AI at exactly two points. Since I lies inside the triangle and I_A lies outside, the point S described in the problem must coincide with I_A . It remains to show $AI_A \cdot AI = AB \cdot AC$.

Compare triangles ABI and $AI_A C$. The collinearity of A , I , I_A on the bisector of $\angle A$ gives $\angle BAI = \angle I_A AC = A/2$. Also, $\angle ABI = B/2$. In the cyclic quadrilateral $IBI_A C$ inscribed in the circle centered at M , the inscribed-angle theorem gives $\angle AI_A C = B/2$ as well. Therefore triangles ABI and $AI_A C$ are similar by AA, and the ratio of corresponding sides gives

$$\frac{AB}{AI_A} = \frac{AI}{AC} \quad \implies \quad AI_A \cdot AI = AB \cdot AC.$$

Since $S = I_A$, we conclude $AS \cdot AI = AB \cdot AC$.

Problem 5

Part (a). Let α be a complex root of P . Setting $x = \alpha$ in the identity $P(x)P(x-1) = P(x^2)$ gives $P(\alpha^2) = 0$. By induction, α^{2^k} is a root for every positive integer k . A nonzero polynomial has only finitely many roots, so this sequence must be finite; this forces $|\alpha| \in \{0, 1\}$.

Setting $x = \alpha + 1$ in the identity produces $P(\alpha + 1)P(\alpha) = P((\alpha + 1)^2)$, and since $P(\alpha) = 0$, we obtain $P((\alpha + 1)^2) = 0$. Thus $(\alpha + 1)^2$ is also a root, and by the same argument $|\alpha + 1| \in \{0, 1\}$.

Part (b). We must identify all complex numbers α satisfying both modulus conditions. If $|\alpha| = 0$ then $\alpha = 0$, and indeed $|0 + 1| = 1$, so $\alpha = 0$ is admissible.

If $|\alpha| = 1$, write $\alpha = e^{i\theta}$. Then

$$|\alpha + 1|^2 = (\cos \theta + 1)^2 + \sin^2 \theta = 2 + 2 \cos \theta,$$

which must equal 0 or 1. The value 0 gives $\cos \theta = -1$, so $\alpha = -1$. The value 1 gives $\cos \theta = -1/2$, so $\alpha \in \{\omega, \omega^2\}$, the primitive cube roots of unity.

The possible roots are therefore $\{0, -1, \omega, \omega^2\}$, and we may write

$$P(x) = c x^a (x + 1)^b (x^2 + x + 1)^d.$$

Substituting this form into the original identity and simplifying the left side gives

$$c^2 x^{a+b} (x-1)^a (x+1)^b (x^2+x+1)^d (x^2-x+1)^d,$$

while the right side equals $c x^{2a} (x^2+1)^b (x^4+x^2+1)^d$. Comparing leading coefficients gives $c = 1$ (the zero polynomial being the other option). Matching the $(x-1)$ factors forces $a = 0$, which then forces $b = 0$. The surviving part $(x^2+x+1)^d (x^2-x+1)^d = (x^4+x^2+1)^d$ matches the right side identically.

Thus the complete list of solutions is $P(x) \equiv 0$ and $P(x) = (x^2 + x + 1)^n$ for any nonnegative integer n .

Problem 6

Part (a). We show by induction that $0 < x_n < 1$ for every n . The base case $x_1 = 1/2$ is immediate. Assume $0 < x_k < 1$. The recurrence rewrites as

$$x_{k+1} = x_k(1 - x_k).$$

Both factors lie strictly in $(0, 1)$, so their product does too, completing the induction.

Next, $x_{n+1} - x_n = -x_n^2 < 0$, so the sequence is strictly decreasing and bounded below by 0. By the Monotone Convergence Theorem it converges to some $L \geq 0$. Taking limits in the recurrence gives $L = L - L^2$, hence $L = 0$.

Part (b). Set $y_n = 1/x_n$. Inverting the recurrence,

$$y_{n+1} = \frac{1}{x_n(1 - x_n)} = \frac{1}{x_n} + \frac{1}{1 - x_n} = y_n + \frac{1}{1 - x_n}.$$

So $y_{n+1} - y_n = \frac{1}{1 - x_n}$.

Since the denominator sequence n is strictly increasing and unbounded, the Stolz–Cesàro theorem gives

$$\lim_{n \rightarrow \infty} \frac{y_n}{n} = \lim_{n \rightarrow \infty} \frac{y_{n+1} - y_n}{1} = \lim_{n \rightarrow \infty} \frac{1}{1 - x_n} = 1.$$

Therefore $\lim_{n \rightarrow \infty} nx_n = 1$.

Part (c). From part (b), $y_{n+1} - y_n = 1 + \frac{x_n}{1 - x_n}$. Telescoping from $k = 1$ to $n - 1$ gives

$$y_n - y_1 = (n - 1) + \sum_{k=1}^{n-1} \frac{x_k}{1 - x_k},$$

and using $y_1 = 2$,

$$y_n = n + 1 + \sum_{k=1}^{n-1} \frac{x_k}{1 - x_k}.$$

Now rewrite the target expression:

$$n - n^2 x_n = n - \frac{n^2}{y_n} = \frac{n(y_n - n)}{y_n} = \frac{n \left(1 + \sum_{k=1}^{n-1} \frac{x_k}{1 - x_k} \right)}{y_n}.$$

Dividing by $\log n$, the target limit becomes the product

$$\left(\lim_{n \rightarrow \infty} \frac{n}{y_n} \right) \cdot \left(\lim_{n \rightarrow \infty} \frac{1 + \sum_{k=1}^{n-1} \frac{x_k}{1 - x_k}}{\log n} \right).$$

The first factor equals 1 by part (b). For the second factor, apply Stolz–Cesàro with denominator $\log n$:

$$\lim_{n \rightarrow \infty} \frac{\frac{x_n}{1 - x_n}}{\log(1 + 1/n)} = \lim_{n \rightarrow \infty} \left(\frac{nx_n}{1 - x_n} \cdot \frac{1/n}{\log(1 + 1/n)} \right).$$

As $n \rightarrow \infty$, we have $nx_n \rightarrow 1$, $1 - x_n \rightarrow 1$, and the standard limit $\frac{1/n}{\log(1 + 1/n)} \rightarrow 1$. The product is 1, and multiplying the two factors together gives

$$\lim_{n \rightarrow \infty} \frac{n - n^2 x_n}{\log n} = 1.$$

Problem 7

The functional equation is $f(x + y) + f(x - y) = 2f(x)f(y)$ for all real x, y , with the additional data that some $a \neq 0$ satisfies $f(a) = 1$ and f is bounded on a neighborhood of 0.

Setting $y = 0$ gives $2f(x) = 2f(x)f(0)$. Since $f(a) = 1 \neq 0$, the function is not identically zero, so $f(0) = 1$. Setting $x = 0$ gives $f(y) + f(-y) = 2f(y)$, so $f(-y) = f(y)$: the function is even.

Setting $y = x$ gives the duplication formula $f(2x) = 2f(x)^2 - 1$. Substituting $u = x + y$, $v = x - y$ transforms the original equation into $f(2x) + f(2y) = 2f(x + y)f(x - y)$, which combined with the duplication formula yields

$$f(x + y)f(x - y) = f(x)^2 + f(y)^2 - 1.$$

Setting $y = a$ in this identity gives $f(x + a)f(x - a) = f(x)^2$. Setting $y = a$ in the original equation gives $f(x + a) + f(x - a) = 2f(x)$. The pair $(f(x + a), f(x - a))$ therefore satisfies a quadratic whose sum is $2f(x)$ and whose product is $f(x)^2$, namely $T^2 - 2f(x)T + f(x)^2 = (T - f(x))^2 = 0$. Hence $f(x + a) = f(x - a) = f(x)$, proving that f is periodic with period a .

Turn now to boundedness. Suppose $|f(x)| \leq M$ on some $(-\delta, \delta)$. The duplication formula $f(2x) = 2f(x)^2 - 1$ extends this bound to $(-2\delta, 2\delta)$, and by induction to $(-2^n\delta, 2^n\delta)$ for all n . Hence f is globally bounded.

Now suppose for contradiction that $|f(x_0)| > 1$ at some x_0 . Write $f(x_0) = \varepsilon \cosh t$ with $t > 0$ and $\varepsilon \in \{+1, -1\}$. The duplication formula gives $f(2x_0) = 2 \cosh^2 t - 1 = \cosh 2t$, and by induction $f(2^n x_0) = \cosh(2^n t)$, which diverges. This contradicts the global bound on f . Hence $|f(x)| \leq 1$ everywhere.

Since f takes values in $[-1, 1]$, we may write $f(x) = \cos(cx)$. Any bounded measurable solution of d'Alembert's equation is continuous, and the complete set of such solutions is $f(x) = \cos(cx)$ for some real constant c . The condition $f(a) = 1$ forces $ca = 2\pi k$ for some integer k , giving $c = 2\pi k/a$. The complete family of solutions is therefore

$$f(x) = \cos\left(\frac{2\pi kx}{a}\right), \quad k \in \mathbb{Z},$$

which includes the constant function $f \equiv 1$ at $k = 0$.

Problem 8

Write x, y, z for a_1, a_2, a_3 , so $x + y + z = 1$ with $x, y, z > 0$. The cyclic sum has a typical term

$$\frac{x}{1-x} \cdot \frac{x}{x+x} \cdot \frac{y}{x+y} \cdot \frac{z}{x+z}.$$

Since $1 - x = y + z$ and the factor $\frac{x}{x+x} = \frac{1}{2}$, the first term simplifies to

$$\frac{x}{y+z} \cdot \frac{1}{2} \cdot \frac{y}{x+y} \cdot \frac{z}{x+z} = \frac{xyz}{2(x+y)(y+z)(z+x)}.$$

By symmetry, the y - and z -terms give the same expression, so the total cyclic sum equals

$$\frac{3xyz}{2(x+y)(y+z)(z+x)}.$$

The bound $\frac{3xyz}{2(x+y)(y+z)(z+x)} \leq \frac{3}{16}$ is equivalent to

$$8xyz \leq (x+y)(y+z)(z+x),$$

which follows from AM-GM applied factor by factor: $x+y \geq 2\sqrt{xy}$, $y+z \geq 2\sqrt{yz}$, $z+x \geq 2\sqrt{zx}$. Multiplying these three inequalities gives exactly $(x+y)(y+z)(z+x) \geq 8xyz$.

Equality holds in AM-GM if and only if $x = y = z$, which combined with $x + y + z = 1$ gives the unique equality case $a_1 = a_2 = a_3 = 1/3$.

End of solutions